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CELL MORPHOLOGY ANALYSIS IN OCULAR IMAGES

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Abstract

This project aims to analyze cell morphology in ocular images using artificial intelligence-based deep learning models. The study involves developing an application designed to accelerate the diagnostic process for medical professionals while reducing errors caused by manual handling. Methods include YOLOv8 for object detection and classification models such as EfficientNet V2, DeiT, and Swin.

The project provides a user-friendly interface capable of analyzing cell morphology and categorizing various types of anomalies. Throughout the research process, multiple datasets and optimization techniques were compared, with the best performance achieved using the RMSprop optimization method.

The findings demonstrate the impact of dataset homogeneity and model selection on performance. This application is expected to serve as a valuable foundation for future studies and improve efficiency in clinical environments.

Keywords: Cell Morphology Analysis, Sperm Cell Detection, YOLO, DeIT, BeIT, Swin

I. Introduction

The continuous advancement of AI systems makes their integration into medical applications inevitable, paving the way for enhanced decision-making and reduced human error in critical tasks and provide assistance for doctors.

This project addresses a significant problem faced by medical professionals: errors or oversight during sperm cell morphology analysis. The application developed in this project serves as a supporting tool, providing enhanced accuracy and efficiency in clinical diagnoses. By automating sperm cell detection and classification in ocular images, the system reduces manual labor and streamlines the diagnostic process for healthcare practitioners.

The primary goal is to create a reliable system that identifies and categorizes sperm cells, aiding doctors in achieving better diagnostic precision. Using object detection and classification techniques, the application is designed to analyze user-uploaded images efficiently and output results with clear visualizations and categorizations.

II. System Design / Sistem Tasarımı

The proposed system is structured into three main modules, each designed to facilitate an efficient and accurate analysis of ocular images. The **User Interface Module** provides users with a web-based platform, developed using React, where they can upload ocular images for analysis. The interface supports multiple file uploads simultaneously, making the workflow more efficient. Once the processing is complete, the results are displayed, highlighting detected sperm cells along with their classifications. Additionally, users have the option to download outputs in structured formats, including YOLO bounding box annotations and classified sperm images, which are organized into folders based on their specific abnormality categories.

The **Detection Module** plays a critical role in identifying sperm cells within the uploaded images. Using the YOLOv8 object detection model, this module processes the images to detect and localize sperm cells. The detected sperm cells are then cropped based on their bounding box locations, after which they are forwarded to the classification module for further analysis.

Within the **Classification Module**, the system performs two stages of classification. Initially, the cropped sperm cell images are analyzed using the DeiT Base model, which categorizes them into four main groups: Normal, Head Dysmorphias, Tail Dysmorphias, and Neck Dysmorphias. If an abnormality is detected, a specialized sub-classification model further identifies specific types of defects. For instance, in cases where a sperm cell is labeled under Head Dysmorphias, it is further classified into subcategories such as Macrocephaly (large head), Microcephaly (small head), Pinhead, or Tapered Head.

In addition to these core modules, **Supporting Components** play a vital role in ensuring the system's reliability and accuracy. **Dataset Preparation** was a key part of the process, involving the use of three different datasets of ocular images for training the object detection model. For classification, datasets containing sperm images were carefully grouped into different categories, with a pseudo-labeling technique applied to reduce manual annotation efforts. To optimize the system's performance, multiple optimization techniques—including RMSprop, Adamax, and SGD—were tested to determine the most effective approach.

From a technical standpoint, the backend of the system is designed to handle computationally intensive processes efficiently. The model training and inference tasks were carried out in a GPU-enabled environment, ensuring fast and accurate processing. The system was implemented using deep learning frameworks such as PyTorch, which provided robust support for model development and performance optimization.

Overall, the system design integrates a seamless workflow between modules, ensuring high efficiency and accuracy in sperm cell detection, classification, and results presentation. Through a combination of deep learning techniques and structured data processing, this approach provides an effective solution for automating cell morphology analysis in ocular images.

Figure 1. Block Diagram.

III. Experimental Results

1. Performance Analysis

The YOLOv8 object detection model was trained using three different datasets of ocular images. Among these datasets, Dataset 2 demonstrated the highest performance due to its balanced and homogeneous sample distribution, achieving a Mean Average Precision (mAP) score surpassing the 75% threshold (Figure 3). For sperm cell classification, multiple deep learning models, including DeiT Base, EfficientNet V2, and BeIT, were evaluated. Among these, the DeiT Base model consistently delivered the best accuracy, achieving an overall classification accuracy of up to 82% on the Boya 1 dataset when optimized with RMSprop (Figure 2). The classification task performed particularly well in detecting Tail Dysmorphias, achieving an F1-score of 0.91, demonstrating strong agreement between true and predicted classifications.

2. Comparison of Optimization Techniques

The study also compared different optimization techniques to determine their impact on model performance. RMSprop consistently outperformed other methods, delivering the most stable and accurate results across all models and datasets. It was followed by Adamax and SGD, both of which provided competitive but slightly lower accuracy. The choice of optimizer played a crucial role in determining the model's ability to generalize effectively across different datasets, further emphasizing the importance of optimization strategies in deep learning-based classification tasks.

En iyi değer	400	800	1600	3200	5995
Train/box loss	1.3056	1.116	0.9893	0.87861	0.82185
Train/cls loss	0.91381	0.82895	0.72191	0.63603	0.60091
Train/dfl loss	1.1851	1.1273	1.0651	1.0183	0.99514
Metrics/ precision(B)	0.80274	0.83633	0.8827	0.89807	0.89461
Metrics/recall(B)	0.77892	804	0.84791	0.86399	0.8762
Metrics/mAP50(B)	0.84771	0.87746	0.92457	0.93994	0.94164
Metrics/ mAP50-95(B)	0.47854	0.57528	0.66653	0.72857	0.75833
Val/box_loss	1.4387	1.1317	0.92929	0.79208	0.70245
Val/cls_loss	0.95053	0.78264	0.65166	0.55979	0.50818
Val/dfl_loss	1.2046	1.1046	1.0227	0.9752	0.93812
Lr/pg0	0.000099	0.000099	0.000099	0.000099	0.000082
Lr/pg1	0.000099	0.000099	0.000099	0.000099	0.000082
Lr/pg2	0.000099	0.000099	0.000099	0.000099	0.000082

Figure 2. Results of Boya 1 Training with Pseudo-Labeling by image number.

Boya_3Class	Optimizasyon	Boya 1	Boya 2	Boya 3
Efficient Net V2 M	Adamax	75%	77%	75%
	RMS Prop	79%	82%	78%
	SGD	77%	80%	76%
BEiT Base	Adamax	65,52%	67,45%	70,65%
	RMS Prop	75,32%	80,04%	71,19%
	SGD	69,83%	77,48%	74,34%
DEiT Base	Adamax	67,12%	73,13%	70,63%
	RMS Prop	78,96%	82,16%	80,70%
	SGD	75,77%	80,36%	78%
Swin	Adamax	72,84%	78,83%	71,80%
	RMS Prop	79,24%	82,36%	80,34%
	SGD	80,17%	79,62%	76,40%

Figure 3. Results of Different Models with different datasets and different optimizers.

IV. Applicatiton Results

When a user accesses the application, the process is designed to be straightforward and user-friendly. The user begins by logging into the application and selecting the images they wish to upload through a file explorer. The system allows for multiple images to be uploaded simultaneously, streamlining the workflow and making the process more efficient.

Once the images are uploaded, they are processed by the YOLOv8 model to detect sperm cells. The detected cells are highlighted with bounding boxes directly on the uploaded image, providing immediate visual feedback to the user. After detection, the cropped sperm cells are analyzed further and categorized into four main groups: Normal, Head Dysmorphias, Tail Dysmorphias, and Neck Dysmorphias. For cells identified as abnormal, the system performs an additional classification to determine the specific defect type, such as macrocephaly or microcephaly.

The results are then displayed on the interface for the user to review. To enhance usability, the application offers options to download the results. Users can obtain YOLO bounding box labels in `.txt` format or download the cropped and classified sperm images. These images are organized into folders based on defect types and subcategories, which are conveniently packaged in a zip file. Buttons can be seen on Figure 4.

This seamless automated process provides precise and categorized results for the uploaded images in minimal time, offering a user-friendly experience suitable for both clinical and research purposes.

This automated process ensures a seamless and efficient experience, providing precise, categorized results for uploaded images in minimal time. (Figure 5)

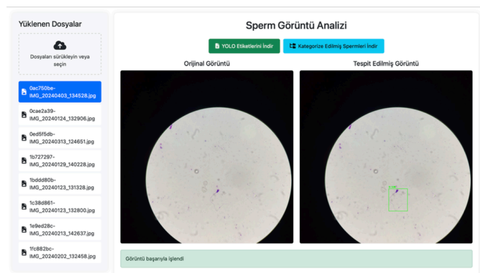

Figure 4. Screen After Uploading a Picture

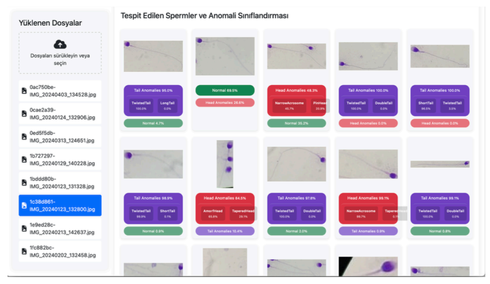

Figure 5. Detailed Classification Result Screen

Conclusion

This study aimed to analyze cell morphology in ocular images by comparing different deep learning methods and optimization techniques. The primary objective was to reduce the workload of medical professionals and improve diagnostic accuracy through automated image processing.

The results demonstrated that RMSprop was the most effective optimization method, consistently outperforming other techniques across all models and datasets. Among the classification models evaluated, DeiT Base provided the most reliable accuracy in three-class classification tasks. Additionally, the Boya 2 dataset yielded the highest performance due to its balanced and homogeneous nature, highlighting the significant role of dataset quality in model performance.

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